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Experimental 1: Creating Class Diagram

Case Study 1:

In a company, one of the data that is processed is employee data. Each employee has an id, name, gender, title, title and salary. Each employee can also view personal data and see his salary.

Employee
+ Id: int + Name: string + Gender: string + Title: string + Salary: int
+ viewPersonalData(): void + seeSalary(): void

1. Describe the class diagram design from case study 1!
The class diagram is a class named employee with 5 attributes and 2 methods
2. Mention what classes can be made from case study 1!
Employee class
3. Mention the attributes and data types that can be identified from each class from case study 1!
Employee class has 5 attributes, which are: id as int, name as string, gender as string, title as string, and salary as int
4. Mention the methods that you created from each class in case study 1!
Employee class has 2 methods, which are: viewPersonalData() and seeSalary()

Experimental 2: Create and accessing the member of a class

Case study 2:

Look at the class diagram below. Make a program based on the class diagram!

Mahasiswa
+nim: int
+nama: String
+alamat: String
+kelas: String
+tampilBiodata(): void

1. Open the text editor/ IDE ex: Notepad ++ / netbeans.
2. Write the code below:

```

1 public class Mahasiswa {
2     public int nim;
3     public String nama;
4     public String alamat;
5     public String kelas;
6
7     public void tampilBiodata() {
8         System.out.println ("Nim      : "+nim);
9         System.out.println ("Nama      : "+nama);
10        System.out.println ("Alamat   : "+alamat);
11        System.out.println ("Kelas   : "+kelas);
12    }
13 }

```

3. Save it into Mahasiswa.java.

The screenshot shows an IDE window titled "week 2 - Class and Object > Mahasiswa.java > Mahasiswa". The code is the same as in the previous block, but with syntax highlighting. The status bar at the bottom indicates "You, 5 minutes ago • week 2 ...".

```

1 public class Mahasiswa {
2     public int nim;
3     public String nama;
4     public String alamat;
5     public String kelas;
6
7     public void tampilBiodata() {
8         System.out.println("Nim      : " + nim);
9         System.out.println("Nama      : " + nama);
10        System.out.println("Alamat   : " + alamat);
11        System.out.println("Kelas   : " + kelas);
12    }
13 }

```

4. To access the members of an object, the instance of the class must be created first. To show how to access the members of the Student class, let's create a new file and then

typing the following program code:

```
1 public class TestMahasiswa {
2     public static void main (String args[]) {
3         Mahasiswa mhs1=new Mahasiswa();
4         mhs1.nim=101;
5         mhs1.nama="Lestari";
6         mhs1.alamat="Jl. Vinolia No 1A";
7         mhs1.kelas="1A";
8         mhs1.tampilBiodata();
9     }
10 }
```

5. Save it into TestMahasiswa.java

ek 2 - Class and Object > TestMahasiswa.java > TestMahasiswa > main(String[])

```
1 public class TestMahasiswa {
2     public static void main(String args[]) {
3         Mahasiswa mhs1 = new Mahasiswa();
4         mhs1.nim = 101;
5         mhs1.nama = "Lestari";
6         mhs1.alamat = "Jl. Vinolia No 1A";
7         mhs1.kelas = "1A";
8         mhs1.tampilBiodata();
9     }
10 }
```

6. Run the TestMahasiswa class

```
PS E:\Pelajaran\POLINEMA\SEM 3\PBO\Practicum> & 'D:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCod
eDetailsInExceptionMessages' '-cp' 'C:\Users\Gamer2Language\AppData\Roaming\Code\User\workspaceStorage\ebd
6d6f861b0ac577c26566366466c73\redhat.java\jdt_ws\Practicum_702e5392\bin' 'TestMahasiswa'
Nim : 101
Nama : Lestari
Alamat : Jl. Vinolia No 1A
Kelas : 1A
PS E:\Pelajaran\POLINEMA\SEM 3\PBO\Practicum>
```

7. Based on the code, please explain in which line the attribute declaration was?
in line 2 – 5

```
1  ✓ public class Mahasiswa {  
2      public int nim;  
3      public String nama;  
4      public String alamat;  
5      public String kelas;  
6  }
```

8. Based on the code, please explain in which line the method declaration was?
line 7

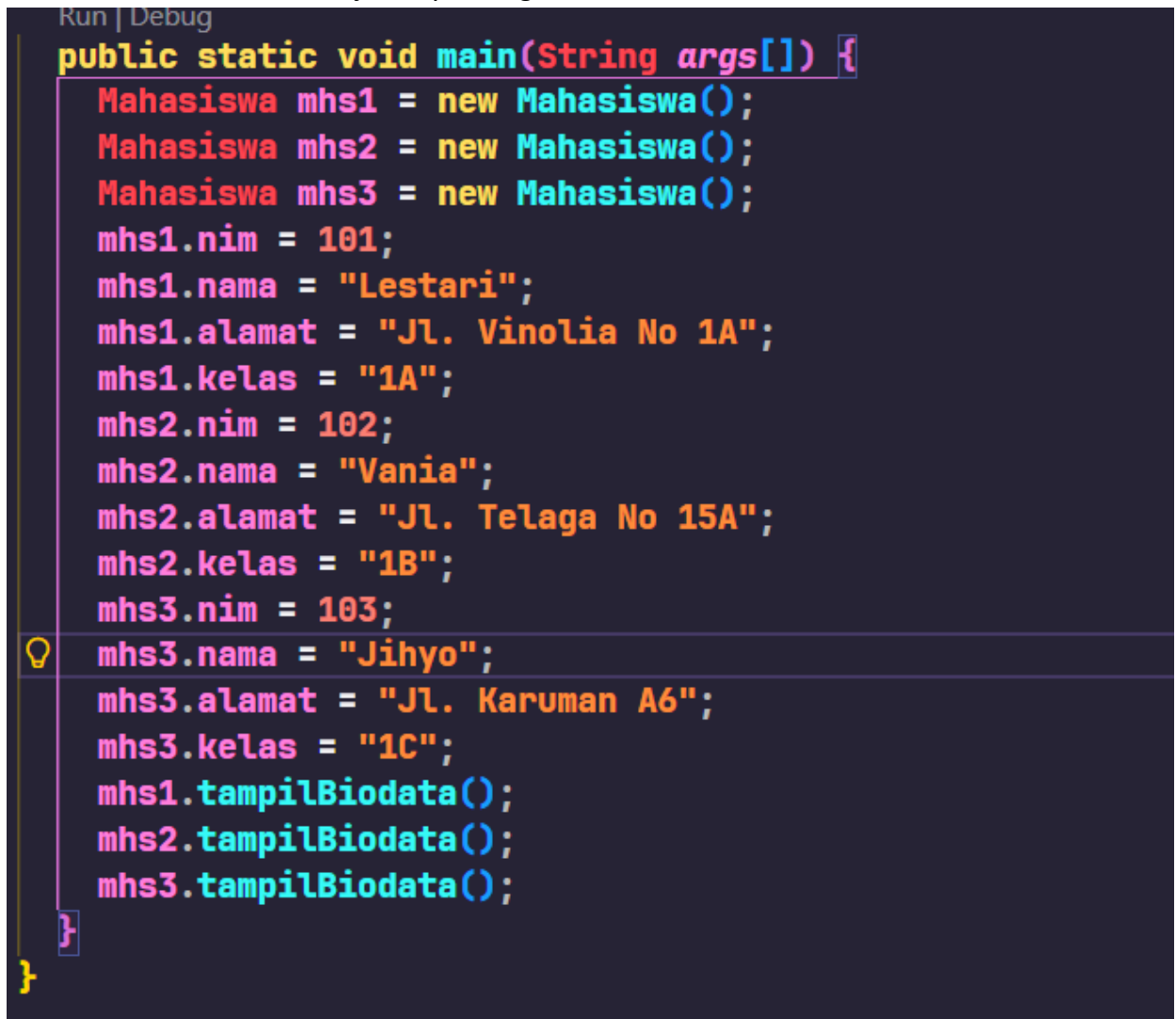
```
7  ✓ public void tampilBiodata() {  
8      System.out.println("Nim      : " + nim);  
9      System.out.println("Nama      : " + nama);  
10     System.out.println("Alamat    : " + alamat);  
11     System.out.println("Kelas    : " + kelas);  
12     }  
13 }
```

9. How many objects instantiate from the code?
once

```
Run | Debug  
public static void main(String args[]) {  
    Mahasiswa mhs1 = new Mahasiswa();  
    mhs1.nim = 101;  
    mhs1.nama = "Lestari";  
    mhs1.alamat = "Jl. Vinolia No 1A";  
    mhs1.kelas = "1A";  
    mhs1.tampilBiodata();  
}
```

10. What does this line “mhs1.nim=101” mean?
assigning the value 101 to the nim of the mhs1 object
11. What does this line “mhs1.tampilBiodata()” do?
Running the tampilBiodata() method inside mhs1 object

12. Please instantiate 2 more object, by adding more code!

A screenshot of a Java IDE with a dark theme. The code is written in a monospaced font with syntax highlighting. The code defines a main method that creates three Mahasiswa objects (mhs1, mhs2, mhs3) and sets their attributes (nim, nama, alamat, kelas). It then calls the tampilBiodata() method for each object. The IDE interface includes a 'Run | Debug' menu at the top left and a lightbulb icon on the left margin next to the mhs3.nama assignment line.

```
Run | Debug
public static void main(String args[]) {
    Mahasiswa mhs1 = new Mahasiswa();
    Mahasiswa mhs2 = new Mahasiswa();
    Mahasiswa mhs3 = new Mahasiswa();
    mhs1.nim = 101;
    mhs1.nama = "Lestari";
    mhs1.alamat = "Jl. Vinolia No 1A";
    mhs1.kelas = "1A";
    mhs2.nim = 102;
    mhs2.nama = "Vania";
    mhs2.alamat = "Jl. Telaga No 15A";
    mhs2.kelas = "1B";
    mhs3.nim = 103;
    mhs3.nama = "Jihyo";
    mhs3.alamat = "Jl. Karuman A6";
    mhs3.kelas = "1C";
    mhs1.tampilBiodata();
    mhs2.tampilBiodata();
    mhs3.tampilBiodata();
}
```

Here I added mhs2 and mhs3 object

Experimental 3: Writing method that has a return value

Scaffolding:

1. Open the text editor/ IDE ex: Notepad ++ / netbeans.

2. Write the code below:

```
1 public class Barang {
2     public String namaBrg;
3     public String jenisBrg;
4     public int stok;
5
6     public void tampilBarang() {
7         System.out.println ("Nama Barang      : "+namaBrg);
8         System.out.println ("Jenis Barang      : "+jenisBrg);
9         System.out.println ("Stok          : "+stok);
10    }
11
12    //method dengan argumen dan nilai balik (return)
13    public int tambahStok(int brgMasuk) {
14        int stokBaru=brgMasuk+stok;
15        return stokBaru;
16    }
17 }
```

3. Save into Barang.java

Week 2 - Class and Object >  Barang.java >  Barang

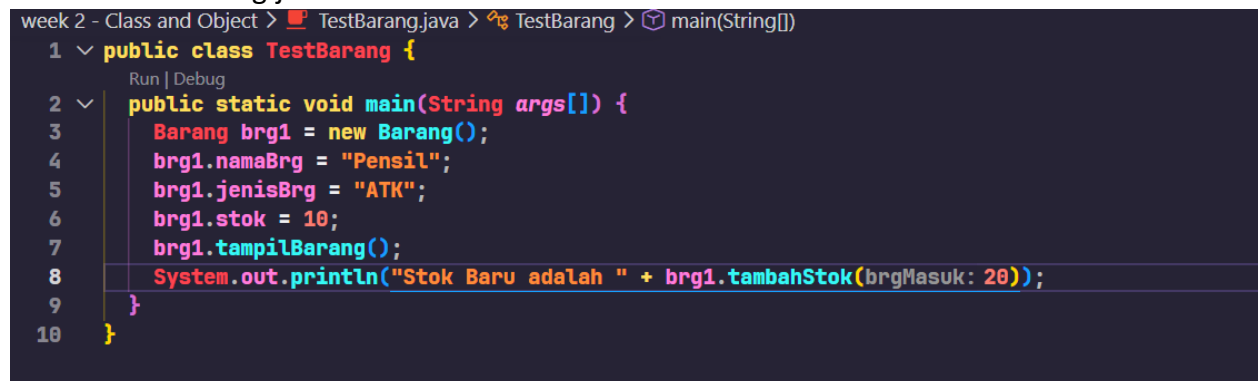
```
1 public class Barang {
2     public String namaBrg;
3     public String jenisBrg;
4     public int stok;
5
6     public void tampilBarang() {
7         System.out.println("Nama Barang      : " + namaBrg);
8         System.out.println("Jenis Barang      : " + jenisBrg);
9         System.out.println("Stok          : " + stok);
10    }
11
12     public int tambahStok(int brgMasuk) {
13         int stokBaru = brgMasuk + stok;
14         return stokBaru;
15     }
16 }
```

1. To access the members of an object, the instance of the class must be created first. To show how to access the members of the barang class, let's create a new file and then

typing the following program code:

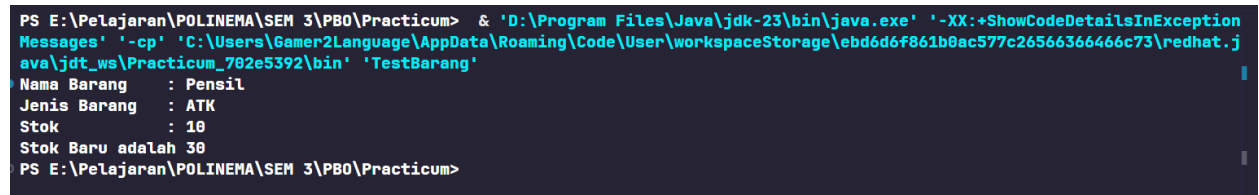
```
1 public class TestBarang{
2     public static void main (String args[]){
3         Barang brg1=new Barang();
4         brg1.namaBrg="Pensil";
5         brg1.jenisBrg="ATK";
6         brg1.stok=10;
7         brg1.tampilBarang();
8         // menampilkan dan mengisi argumen untuk menambahkan stok barang
9         System.out.println ("Stok Baru adalah " +brg1.tambahStok(20));
10    }
11 }
```

2. Save into TestBarang.java



```
week 2 - Class and Object > TestBarang.java > TestBarang > main(String[])
1 public class TestBarang {
2     public static void main(String args[]) {
3         Barang brg1 = new Barang();
4         brg1.namaBrg = "Pensil";
5         brg1.jenisBrg = "ATK";
6         brg1.stok = 10;
7         brg1.tampilBarang();
8         System.out.println("Stok Baru adalah " + brg1.tambahStok(brgMasuk: 20));
9     }
10 }
```

3. Run the code!



```
PS E:\Pelajaran\POLINEMA\SEM 3\PB0\Practicum> & 'D:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Gamer2\Language\AppData\Roaming\Code\User\workspaceStorage\ebd6d6f861b0ac577c26566366466c73\redhat.j
ava\jdt_ws\Practicum_702e5392\bin' 'TestBarang'
Nama Barang : Pensil
Jenis Barang : ATK
Stok : 10
Stok Baru adalah 30
PS E:\Pelajaran\POLINEMA\SEM 3\PB0\Practicum>
```

4. What is the function of an argument in a method?

to pass a value to the method that we are calling

5. Makes conclusion on “return” keyword, when should we used it?

we should use it when the method needs to return a value

4.2 Assignments

1. One of the video game rental shops process is borrowing, The stored data when someone renting the game are the id, member name, game name, and the amount to pay. Each rent can display the data and the amount to pay. Make a class diagram based on the case study!

explanation:

- The price amount will be coming from price per day times renting duration!
- Assume that 1 transaction will only consist of 1 game.

Transaction
+ id: int
+ memberName: string
+ gameName: String
- pricePerDay: int
+ rentDuration: int
+ total: int
+ printInfo(): void
+ printTotal(): void

2. Create the code based on the case study no 1!

```

week 2 - Class and Object > Transaction.java > Transaction > total
1  public class Transaction {
2      public int id;
3      public String memberName;
4      public String gameName;
5      private int pricePerDay;
6      public int rentDuration;
7      public int total;
8
9      public void setPricePerDay(int pricePerDay) {
10         this.pricePerDay = pricePerDay;
11     }
12
13     public void printInfo() {
14         System.out.println("Transaction ID : " + id);
15         System.out.println("Member Name : " + memberName);
16         System.out.println("Game Name : " + gameName);
17         System.out.println("Price Per Day : " + pricePerDay);
18         System.out.println("Rent Duration : " + rentDuration + " day(s)");
19     }
20
21     public void printTotal() {
22         this.total = this.rentDuration * this.pricePerDay;
23         System.out.println("Total Price : " + this.total);
24     }
25 }

```

3. More exercise, please create the code from the following class diagram:

Lingkaran
+phi: double
+r: double
+hitungLuas(): double
+hitungKeliling(): double

```
public class Lingkaran {
```

```

    public double phi = 3.14;
    public double r;

    public double hitungLuas() {
        return phi * r * r;
    }

    public double hitungKeliling() {
        return 2 * phi * r;
    }
}

```

4. More exercise, please create the code from the following class diagram:



Description:

- The hargaDasar attribute is in Rupiah and the discount attribute in percentage!
- The hitungHargaJual() method used to calculate the based price by following this rule
 Selling price (hargaJual)= base price(hargaDasar) – (discount(diskon) x base price (hargaDasar))
- The tampilData() Method will be used to show the value of each class attribute: kode, namaBarang, hargaDasar, diskon dan harga jual.

```

public class Barang {
    public String kode;
    public String namaBarang;
    public int hargaDasar;
    public float diskon;

    public int hitungHargaJual() {
        return (int) (hargaDasar - ((diskon / 100) * hargaDasar));
    }

    public void tampilData() {
        System.out.println("Kode Barang : " + kode);
        System.out.println("Nama Barang : " + namaBarang);
        System.out.println("Harga Dasar : Rp " + hargaDasar);
    }
}

```

```
System.out.println("Diskon    : " + diskon + "%");  
System.out.println("Harga Jual : Rp " + hitungHargaJual());  
}  
}
```